Università degli Studi di Napoli
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Dipartimento di Ingegneria Elettrica e delle Tecnologie dell'Informazione

Statistical Learning and Data Analysis – Module A

# INTRODUCTION TO PROBABILITY

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## Descriptive vs Inferential Statistics

### Descriptive Statistics (Direct Problem):

- The entire population is observed and summarized through statistical indicators.
- Not always possible or convenient to observe all units.
  - High cost, time, or destruction of observed units.
  - Infinite population.

### Statistical Inference (Indirect Problem):

- Based on a sample to infer characteristics of the population.

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# Basic Definitions

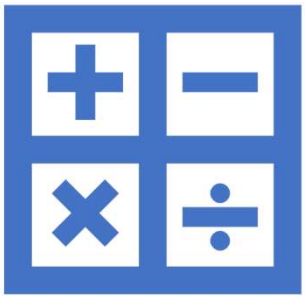
**Population**: the entire group under study.

**Sample**: subset of the population that is observed.

**Statistical inference**: process of drawing conclusions about the population from data collected on a sample, using probability reasoning.



## Types of Inferential Problems



### Estimation of Parameters:

- Point estimation → assign a single value to the parameter.
- Interval estimation → assign an interval of plausible values.

### Hypothesis Testing

- Decide whether a hypothesis on a parameter is acceptable in probabilistic terms.

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# Introductory Concepts

Not all variables are observed for every unit of the population.

- Observations are made only on a **sample**.

## Main questions:

- How to extend sample conclusions to the population?
- What are the limits of validity and reliability?

**Solution**: use an **unbiased random sample** — units not influenced by the studied variable.

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# The Sample

Observations are chosen by **random selection** (random draw).

- If the survey is not random, inferential methods cannot be applied.
- Use an **unbiased sample**: units unaffected by the variable under study.
- Best approach → random draw of sampling units.

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# Random Sample

**Definition:** a sample selected by random draw (random selection).

- Each unit has the same probability of being chosen.
- Ensures representativeness and validity of inference.
- Non-random samples → inferential analysis not applicable.

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# A thought

Probability is the language of Uncertainty.

A question arises and the answer is "maybe".

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# The need

A data scientist uses inferential statistics to make statement about a population based on information contained in a sample of units selected from that population.

Descriptive statistics can be used to summarize the information about the sample. However, a sample is not identical to the population from which it was selected.

We need to assess the degree of accuracy to which the sample mean, sample standard deviation, or sample proportion represent the corresponding population values.

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# How to define probability?

Observations of a phenomenon can result in many different outcome, some are more likely to occur than others.

Several attempts have been made to give a precise definition for the probability of an outcome.

***Classical School***
***Frequentists School***
***Subjectivists School***

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# Classical School

- From **games of chance**, typical probability statements of this type are, for example, *"the probability that a flip of a balanced coin will show 'heads' is* $1/2$*"* and *"the probability of drawing an ace when a single card is drawn from a standard deck of* $52$ *cards is* $4/52$*."* A coin flip has two possible outcomes (head or tail); *the probability of a head should then be* $1/2$. Similarly, there are $4$ aces in a standard deck of $52$ cards, so *the probability of drawing an ace in a single draw is* $4/52$*, or* $4$ *out of* $52$.
- In the classical interpretation of probability, each possible distinct result is an outcome; an event is identified as a collection of outcomes.
- The probability of an event $E$ under the classical interpretation of probability is computed by taking the ratio of the number $n_E$ of outcomes favorable to event E to the total number $n$ of possible outcomes:

$$P(E) = \frac{n_E}{n}$$

- *The applicability of this interpretation depends on the assumption that all outcomes are equally likely.* If this assumption does not hold, the probabilities indicated by the classical interpretation of probability will be in error.

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# Frequentists School

- It is an **empirical approach**.
- If an experiment is repeated a large number of times and the event $E$ occurs $30\%$ of the time, then $0.3$ should be a very good approximation to the probability of event E. That is, if an experiment is conducted $n$ different times and if the event $E$ occurs on $n_E$ of these trials, then the probability of event $E$ is approximately:

$$\lim_{n\to\infty} \frac{n_E}{n} = p_E$$

- We say "approximate" because we think of the actual probability $P(E)$ as the relative frequency of the occurrence of event $E$ over a very large number of observations or repetitions of the phenomenon.
- *Not all experiments can be repeated!!!*

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# Subjectivists School

- **Subjective interpretation:** This interpretation of probability can be used for problems in which it is difficult to imagine a repetition of an experiment. *It derives from an individual's personal judgment or own experience about whether a specific outcome is likely to occur.* It contains no formal calculations and only reflects the subject's opinions and past experience rather than on data or computation.
- An individual is asked the probability of a dice roll yielding a $6$. The individual looks at the past three rolls and notes that $6$ came up in all instances. The individual believes that the probability of the next dice roll yielding a $6$ is at $30\%$. Although the mathematical prediction is incorrect, the individual's personal experience of the dice roll yielding $6$ in three instances created a situation where he used subjective probability.
- The probability is the odds I am willing to bet on the success of an event to receive $1$ in return. In that respect, the probability is ranging between $0$ and $1$.
- *Subjective probabilities differ from person to person and contain a high degree of personal bias.*

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# Finding the Probability of an Event

- We will use the classical interpretation and the relative frequency concept to illustrate how to compute the probability of an outcome or a simple event.
- Consider an experiment that consists of tossing two coins, a penny and then a dime, and observing the upturned faces. There are four possible outcomes:
  - TT: tails for both coins
  - TH: a tail for the penny, a head for the dime
  - HT: a head for the penny, a tail for the dime
  - HH: heads for both coins
- What is the probability of observing the event exactly one head from the two coins?

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# The answer is …

- The probability can be obtained by assuming that all four outcomes are equally likely.
- In this case, there are $n = 4$ possible outcomes, and $n_E = 2$ of these are favorable for the event of interest, observing exactly one head.
- Hence, by the classical interpretation of probability

$P(One\ Head) = \frac{2}{4} = \frac{1}{2}$

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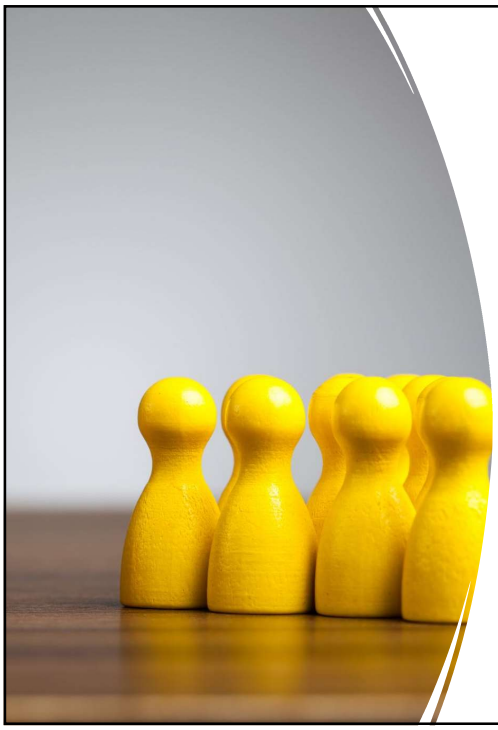

# Probability whatelse

- Probability cannot be defined.
- **Probability** is a primitive concept together with **Event** and **Trial**, which do not need a definition.

*"A **trial** yields an **event** with a given **probability**"*

- This statement includes three primitive concepts of the **axiomatic probability paradigm**, as an alternative to other paradigms (classical, frequentist, subjective).
- We fix Axioms and Fundamental Theorems derived from the axioms.

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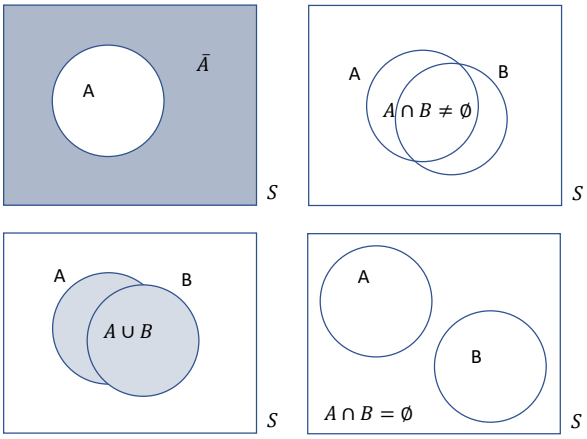
# Venn diagram

The set of all possible outcomes of an experiment is called the **sample space** and it is usually denoted by the letter $S$.

Each outcome in a sample space is called an element of the sample space, or simply a sample point.

We are interested in events that are actually combinations of two or more events, formed by taking **unions**, **intersections**, and **complements**.

Sample spaces and events, particularly relationships among events, are often depicted by means of **Venn diagrams,** in which the sample space is represented by a rectangle, while events are represented by regions within the rectangle, usually by circles or parts of circles.




Two events having no elements in common are said to be **mutually exclusive**. $\emptyset$ is the empty set

Use Boolean Logic Operators on the Events and Visualize the result.

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# Kolmogorov's Axioms

1. **Non-negative real number**: $P(E) \geq 0$, with $P(E) \in \mathcal{R}, \forall E \in S$, being $S$ the sample space.
2. **Unit measure**: $P(S) = 1$, the probability that at least one of the elementary events in the entire sample space will occur is 1, thus $S$ is the event that certainly occurs.
3. **Additivity**: Any countable sequence of *mutually exclusive events* $E_1, E_2, \ldots$ (or disjoint sets) satisfies: $P(\bigcup_{i=1}^{\infty} E_i) = \sum_{i=1}^{\infty} P(E_i)$.
   - If $A \cap B = \emptyset$ (***incompatible events***) then $P(A \cup B) = P(A) + P(B)$

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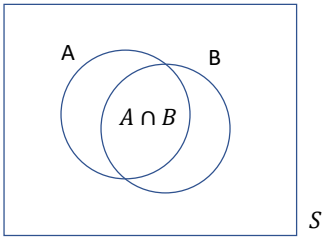
# Fundamental Theorems

- ***The probability of the empty set***: $P(\emptyset) = 0$.
- ***The complement rule***: $P(\bar{E}) = 1 - P(E)$, where $\bar{E}$ is that the event $E$ does not occur.
  - It derives that $0 \leq P(E) \leq 1$.
- ***Additivity Law***: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.

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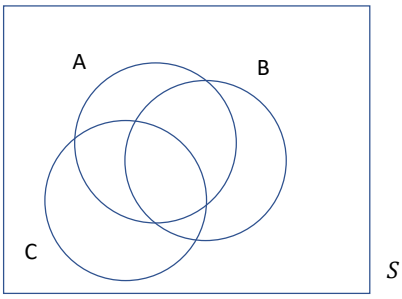
# What about the Additive Law for three events?

- ***Additivity Law***: $P(A \cup B \cup C) = ?$

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# Conditional Probability

- The probability of an event occurring given that another event has already occurred:

$P(B|A) = \frac{P(A \cap B)}{P(A)}$ where $P(A) \neq 0$.

- **Factorization Theorem**:

$$P(B) = P(A \cap B) + P(\bar{A} \cap B) = P(B|A)P(A) + P(B|\bar{A})P(\bar{A})$$

- **Stochastic Independence**:

$$P(A \cap B) = P(A) \times P(B)$$
$$P(B|A) = P(B)$$

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# Organize computation in a two-way table

| Use Glasses? / Gender | $\bar{B}$ | $B$ | *Total* |
|---|---|---|---|
| $\bar{A}$ | $P(\bar{A} \cap \bar{B}) = 0.2$ | $P(\bar{A} \cap B) = 0.4$ | $P(\bar{A}) = 0.6$ |
| $A$ | $P(A \cap \bar{B}) = 0.2$ | $P(A \cap B) = 0.20$ | $P(A) = 0.4$ |
| *Total* | $P(\bar{B}) = 0.4$ | $P(B) = 0.6$ | $P(S) = 1$ |

Sample data

| Use Glasses? / Gender | NO | YES | *Total* |
|---|---|---|---|
| Male | 20 | 40 | *60* |
| Female | 20 | 20 | *40* |
| *Total* | *40* | *60* | **100** |

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## Challenging Questions

- Consider an experiment that consists of tossing two dices, each with possible outcomes 1, 2, 3, 4, 5, 6, each outcome of any dice is equally likely.
- The Maximum face between the two dices has possible outcomes 1, 2, 3, 4, 5, 6.
- The Sum of the two faces of the dices has possible outcomes 2, 3, 4, ….12.

➢ If you have to bet on what is the most likely Maximum face, which outcome do you choose?

➢ If you have to bet on what is the most likely Sum, which outcome do you choose?

➢ Assume that you know that the Maximum face is "4", what is the probability that the Sum is "7"? Is more or less likey when you assume that the Maximum face was "3"?

➢ One of the dice provides the outcome "4", what is the most likely Sum to bet on?

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## Sample space

- The sample space can be derived as follow:

$$\{1,1\}, \{1,2\}, \{1,3\}, \{1,4\}, \{1,5\}, \{1,6\},$$
$$\{2,1\}, \{2,2\}, \{2,3\}, \{2,4\}, \{2,5\}, \{2,6\},$$
$$\{3,1\}, \{3,2\}, \{3,3\}, \{3,4\}, \{3,5\}, \{3,6\},$$
$$\{4,1\}, \{4,2\}, \{4,3\}, \{4,4\}, \{4,5\}, \{4,6\},$$
$$\{5,1\}, \{5,2\}, \{5,3\}, \{5,4\}, \{5,5\}, \{5,6\},$$
$$\{6,1\}, \{6,2\}, \{6,3\}, \{6,4\}, \{6,5\}, \{6,6\}$$

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# A bivariate probability distribution

- The Maximum face between the two dices has possible outcomes 1, 2, 3, 4, 5, 6. These are mutually exclusive events $M_1$, $M_2, \dots, M_6$.
- The Sum of the two faces of the dices has possible outcomes 2, 3, 4, ….12. These are mutually exclusive events $S_2, S_3, \dots, S_{12}$.
- You can organize all probability computations using a bivariate probability distribution of the type $P(M_i \cap S_j)$ for $i = 1, \dots, 6; j = 2, \dots, 12$. You can also derive the row and the column marginal probability distributions, as also the conditional probability distributions.
- Answer to the challenging questions!

| Sum Max | $S_2$ | $S_3$ | $S_4$ | $S_5$ | $S_6$ | $S_7$ | $S_8$ | $S_9$ | $S_{10}$ | $S_{11}$ | $S_{12}$ | *Total* |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| $M_1$ | 1/36 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 1/36 |
| $M_2$ | 0 | 2/36 | 1/36 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 3/36 |
| $M_3$ | 0 | 0 | 2/36 | 2/36 | 1/36 | 2/36 | 0 | 0 | 0 | 0 | 0 | 5/36 |
| $M_4$ | 0 | 0 | 0 | 2/36 | 2/36 | 2/36 | 1/36 | 0 | 0 | 0 | 0 | 7/36 |
| $M_5$ | 0 | 0 | 0 | 0 | 2/36 | 2/36 | 2/36 | 2/36 | 1/36 | 0 | 0 | 9/36 |
| $M_6$ | 0 | 0 | 0 | 0 | 0 | 0 | 2/36 | 2/36 | 2/36 | 2/36 | 1/36 | 11/36 |
| *Total* | 1/36 | 2/36 | 3/36 | 4/36 | 5/36 | 6/36 | 5/36 | 4/36 | 3/36 | 2/36 | 1/36 | 1 |

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# Bayes Theorem

- Suppose that there are only two mutually exclusive events (called **states of nature**), namely $A$ and $\bar{A}$, that can generate the event $E$.
- We know
  - the ***a-priori*** or ***prior probabilities*** of the causes: $P(A)$ and $P(\bar{A})$ such that $P(A) + P(\bar{A}) = 1$;
  - the conditional probabilities that the event $E$ occurs given each of the two causes, named ***likelihoods or probative probabilities***: $P(E|A)$ and $P(E|\bar{A})$.
- Bayes theorem calculates the **posterior conditional probability** that the cause $A$ has generated the event $E$:

$$P(A|E) = \frac{P(A)P(E|A)}{P(E)} = \frac{P(A)P(E|A)}{P(E|A)P(A)+P(E|\bar{A})P(\bar{A})}$$

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# Bayes Theorem general formula

- There is a set of $K$ mutually exclusive *underlying* events (states of nature) $A_k$ with unconditional a-priori or prior probabilities $P(A_k)$.
- There is a set of mutually exclusive *observable* events $E_j$ with likelihoods or probative probabilities $P(E_j|A_k)$.
- The posterior probability $P(A_k|E_j)$ can be calculated as follows:

$$P(A_k|E_j) = \frac{P(A_k)P(E_j|A_k)}{P(E_j)} = \frac{P(A_k)P(E_j|A_k)}{\sum_{k=1}^{K} P(A_k)P(E_j|A_k)}$$

- Prior and posterior refer to probabilities before or after observing an event $E_j$.

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# Combinatorial Calculus

| | Extraction *without replications* | Extraction *with replications* |
|---|---|---|
| Ordered sequence | $D_{n,k} = \frac{P_n}{(n-k)!} = \frac{n!}{(n-k)!}$ | $D^r_{n,k} = n^k$ |
| Non-ordered sequence | $C_{n,k} = \binom{n}{k} = \frac{n!}{k!(n-k)!}$ | $C^r_{n,k} = \binom{n+k-1}{k} = \frac{(n+k-1)!}{k!(n-1)!}$ |

Partition of n objects in k classes $\quad R_{n,k_1,\ldots,k_m} = \frac{n!}{k_1!\ldots k_m!}, \quad \sum_i k_i = n$

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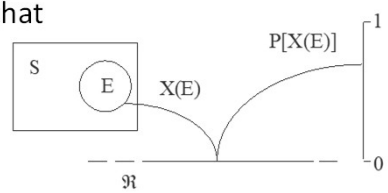
| Events | r.v. $X$ | $P(X)$ |
|---|---|---|
| $E_1$ | $x_1$ | $p_1$ |
| … | … | … |
| $E_i$ | $x_i$ | $p_i$ |
| … | … | … |
| $E_k$ | $x_k$ | $p_k$ |
| | | 1 |

# Discrete Random Variable

- Let $E_1, \dots, E_i, \dots E_k$ be a set of elementary events such that
  - $E_i \cap E_j = \emptyset \; i \neq j$ (mutually exclusive)
  - $\bigcup_{i=1}^{k} E_i = S$ (necessary).
- A discrete random variable $X$ associates a value $x_i$ in real space to each event $E_i$ in a specified finite or countable list of values, with a *probability distribution function* such that
  - $p_i \geq 0$
  - $\sum_{i=1}^{k} p_i = 1.$

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# Fundamental Moments

- $E(X) = \sum_{i=1}^{k} x_i p_i = \mu$ (moment of order 1)
- $E(X^2) = \sum_{i=1}^{k} x_i^2 p_i = \mu_2$ (moment of order 2)
- $var(X) = E[X - E(X)]^2 = \sum_{i=1}^{k} (x_i - \mu)^2 p_i = \mu_2 - \mu^2 = \bar{\mu}_2$ (centered moment of order 2)
- Standard deviation is the square root of the variance.

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# Discrete probability models

- **Uniform** $X \sim U_d(k)$ describes the probability that a finite number $k$ of values are *equally likely* to be observed (i.e., throw of the dice): $P(X = x) = 1/k \quad for\ x = 1, \dots, k$.
- **Bernoulli** $X \sim B(1, \pi)$ describes the probability of either *"success" or "failure"* (i.e., throw of the coin): $P(X = x) = \pi^x (1 - \pi)^{1-x} for\ x = 0,1$.
- **Binomial** $X \sim B(n, \pi)$ describes the probability that a *given number of "successes"* occurs in a fixed number $n$ of *independent* Bernoulli trials, each with the same probability $\pi$: $P(X = x) = \binom{n}{k} \pi^x (1 - \pi)^{n-x} for\ x = 0,1, \dots, n$.
- **Geometric** $X \sim G(\pi)$ describes the probability that a *given number of trials* occurs in a sequence of *independent* Bernoulli trials *before a specified success occurs*: $P(X = x) = \pi^x (1 - \pi)^{x-1} for\ x = 1, 2, \dots$
- **Pascal (Negative Binomial)** $X \sim NB(k, \pi)$ describes the probability that a *given number of trials* occurs in a sequence of *independent* Bernoulli trials (with the same probability $\pi$) before *a specified (non-random) number $k$ of successes occur*.
- **Poisson (law of small numbers)** $X \sim P(\lambda)$ describes the probability that a given number of events occurs *("arrivals")* in a fixed interval of time and/or space, with a known average rate $\lambda$ and independently of the time since the last event (i.e., reliability).
- **Hypergeometric** $X \sim H(N, n, m)$ describes the probability that a *given number of "successes"* occurs in a fixed number of *(dependent)* Bernoulli trials $n$ drawn *without replacement* from a finite population of size $N$ containing $m$ successes.

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# Discrete probability models in a nutshell

| Random Variable | Probability Distribution | Expectation | Variance |
|---|---|---|---|
| Uniform $X \sim U_d(k)$ | $P(X = x) = 1/k \quad for\ x = 1, \dots, k$ | $E(X) = \frac{k+1}{2}$ | $Var(X) = \frac{(k^2 - 1)}{12}$ |
| Bernoulli $X \sim B(1, \pi)$ | $P(X = x) = \pi^x (1 - \pi)^{1-x} \quad for\ x = 0,1$ | $E(X) = \pi$ | $Var(X) = \pi(1 - \pi)$ |
| Binomial $X \sim B(n, \pi)$ | $P(X = x) = \binom{n}{x} \pi^x (1 - \pi)^{1-x}$ $for\ x = 0,1, \dots, n$ | $E(X) = n\pi$ | $Var(X) = n\pi(1 - \pi)$ |
| Pascal or Negative Binomial $X \sim NB(k, \pi)$ | $P(X = x) = \binom{x-1}{k-1} \pi^k (1 - \pi)^{x-k}$ $for\ x = k, k+1, \dots$ | $E(X) = k\frac{1}{\pi}$ | $Var(X) = k\frac{1-\pi}{\pi^2}$ |
| Poisson $X \sim P(\lambda)$ | $P(X = x) = \frac{\lambda^x}{x!} e^{-\lambda}$ $for\ x = 0,1, \dots$ | $E(X) = \lambda$ | $Var(X) = \lambda$ |
| Hypergeometric $X \sim H(N, n, m)$ | $P(X = x) = \frac{\binom{m}{x}\binom{N-m}{n-x}}{\binom{N}{n}}$ $for\ \max(0, n - N + m) \le x \le \min(n, m)$ | $E(X) = n\frac{m}{N}$ | $Var(X) = n\frac{m}{N}\frac{N-n}{N-1}$ |

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# Continuous Random Variable

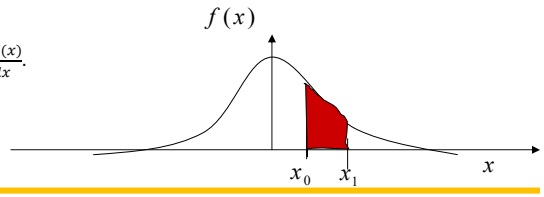
- A continuous random variable $X$ takes any real value $x$ with the **density probability function** $f(x)$ such that
  - $f(x) \geq 0$ for $-\infty \leq x \leq +\infty$
  - $\int_{-\infty}^{+\infty} f(x)dx = 1$
- The **(Cumulative) Distribution Function** is defined as:

$F(x) = P(X \leq x) = \int_{-\infty}^{x} f(x)dx$ for $-\infty \leq x \leq +\infty$

- If $f(x)$ and $F(x)$ are the values of the probability density and the distribution function of a continuous r.v. $X$ at $x$, then

$P(x_0 \leq X \leq x_1) = P(X \leq x_1) - P(X \leq x_0) = F(x_1) - F(x_0)$

for any real constants $x_0$ and $x_1$ such that $x_0 \leq x_1$ and $f(x) = \frac{dF(x)}{dx}$.

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# Fundamental Moments

- $E(X) = \int_{-\infty}^{+\infty} xf(x)dx = \mu$(moment of order 1)
- $E(X^2) = \int_{-\infty}^{+\infty} x^2 f(x)dx = \mu_2$(moment of order 2)
- $var(X) = E[X - E(X)]^2 = \int_{-\infty}^{+\infty} (x-\mu)^2 f(x)dx = \mu_2 - \mu^2 = \bar{\mu}_2$ (centered moment of order 2)
- Standard deviation is the square root of the variance.

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# Continuous probability models

- **Uniform** $X \sim U_c\ (\alpha, \beta)$ describes the probability that all intervals of the same length on a given support are equally likely.
- **Normal** $X \sim N(\mu, \sigma^2)$ arises from the **central limit theorem**, to be the *distribution limit* of the sum of $n$ independent and identically (unknown) distributed random variables, each with mean $\mu$ and variance $\sigma^2$; it is a symmetric distribution where probability intervals can be easily derived from the probability tables of its standardized random variable version which is also normal.
- **Exponential** $X \sim Exp(\lambda)$ describes the probability of a *given time* between events in a Poisson process.
- **Gamma** describes the probability in *life testing*, namely *the waiting time until death.*
- **Beta** describes the probability of *intervals occurring into an interval zero, one.*

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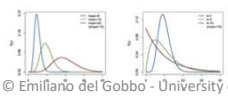
# Continuous probability models

| Random Variable | Probability Density Function | Expectation | Variance |
|---|---|---|---|
| Uniform $X \sim U_c\ (\alpha, \beta)$ | $f(x) = \begin{cases} \frac{1}{\beta - \alpha} & for\ x \in [\alpha, \beta] \\ 0 & otherwise \end{cases}$ | $E(X) = \frac{\alpha + \beta}{2}$ | $Var(X) = \frac{(\beta - \alpha)^2}{12}$ |
| Gaussian or Normal $X \sim N(\mu, \sigma^2)$ | $f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} exp\left\{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2\right\}$ $for\ -\infty < x < +\infty$ | $E(X) = \mu$ | $Var(X) = \sigma^2$ |
| Exponential $X \sim En(\lambda)$ | $f(x) = \lambda\ exp(-\lambda x)$ $for\ 0 \le x < +\infty$ | $E(X) = \frac{1}{\lambda}$ | $Var(X) = \frac{1}{\lambda^2}$ |
| Gamma $X \sim Ga(\kappa, \lambda)$ | $f(x) = \frac{\lambda^\kappa}{\Gamma(\kappa)} x^{\kappa-1} exp(-\lambda x)\ for\ x > 0$ | $E(X) = \frac{\kappa}{\lambda}$ | $Var(X) = \frac{\kappa}{\lambda^2}$ |
| Beta $X \sim Be(\alpha, \beta)$ | $f(x) = \frac{1}{B(\alpha, \beta)} x^{\alpha-1}(1-x)^{\beta-1}$ $for\ 0 \le x \le 1$ | $E(X) = \frac{\alpha}{\alpha + \beta}$ | $Var(X) = \frac{\alpha\ \beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$ |

Weibull $\left(\frac{x}{\lambda}\right)^\beta \sim En(1)\ \lambda > 0, \beta > 0$



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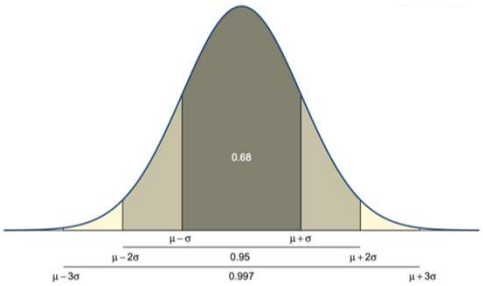
# Normal Distribution

- The density probability distribution of the **Normal Probability Model** is bell-shaped, symmetric with respect to the mean $\mu$ and its form depends on the standard deviation $\sigma$.
- The probability of falling within $\mu \pm \sigma$ is 68.27%, within $\mu \pm 2\sigma$ is 95.45% and within $\mu \pm 3\sigma$ is 99.73%.

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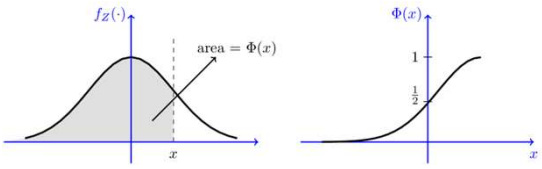
# Standardized Normal Distribution

- Probability intervals of any normal distribution $X \sim N(\mu, \sigma^2)$ can be easily derived from the probability tables of its **standardized random variable which is also normally distributed.**
  - If $X \sim N(\mu, \sigma^2)$ then $Z = \frac{X - \mu_X}{\sigma_X} \sim N(0,1)$.
  - Cumulative Distribution Function (c.d.f.): $P(X \leq x) = P(Z \leq z_x) = F(z)$

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# Transformations of Normal random variable

**Chi-square with g degrees of freedom:**

- The sum of g independent squared standardized normal distributions;

**F-Snedecor Fisher with g and h degrees of freedom:**

- The ratio between two independent chi-square distributions, each divided by its degrees of freedom;

**t-Student with g degrees of freedom:**

- The ratio between a standardized normal distribution and the square root of the chi-square distribution divided by its degrees of freedom, the two r.v. to be independent.

**These transformations are used in inferential testing procedures.**

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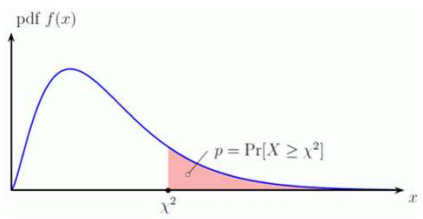
# Chi-square Pdf and Probability Table




https://i.stack.imgur.com/TnMv3.png

**Chi-square Distribution Table**

| d.f. | .995 | .99 | .975 | .95 | .9 | .1 | .05 | .025 | .01 |
|---|---|---|---|---|---|---|---|---|---|
| 1 | 0.00 | 0.00 | 0.00 | 0.00 | 0.02 | 2.71 | 3.84 | 5.02 | 6.63 |
| 2 | 0.01 | 0.02 | 0.05 | 0.10 | 0.21 | 4.61 | 5.99 | 7.38 | 9.21 |
| 3 | 0.07 | 0.11 | 0.22 | 0.35 | 0.58 | 6.25 | 7.81 | 9.35 | 11.34 |
| 4 | 0.21 | 0.30 | 0.48 | 0.71 | 1.06 | 7.78 | 9.49 | 11.14 | 13.28 |
| 5 | 0.41 | 0.55 | 0.83 | 1.15 | 1.61 | 9.24 | 11.07 | 12.83 | 15.09 |
| 6 | 0.68 | 0.87 | 1.24 | 1.64 | 2.20 | 10.64 | 12.59 | 14.45 | 16.81 |
| 7 | 0.99 | 1.24 | 1.69 | 2.17 | 2.83 | 12.02 | 14.07 | 16.01 | 18.48 |
| 8 | 1.34 | 1.65 | 2.18 | 2.73 | 3.49 | 13.36 | 15.51 | 17.53 | 20.09 |
| 9 | 1.73 | 2.09 | 2.70 | 3.33 | 4.17 | 14.68 | 16.92 | 19.02 | 21.67 |
| 10 | 2.16 | 2.56 | 3.25 | 3.94 | 4.87 | 15.99 | 18.31 | 20.48 | 23.21 |
| 11 | 2.60 | 3.05 | 3.82 | 4.57 | 5.58 | 17.28 | 19.68 | 21.92 | 24.72 |
| 12 | 3.07 | 3.57 | 4.40 | 5.23 | 6.30 | 18.55 | 21.03 | 23.34 | 26.22 |
| 13 | 3.57 | 4.11 | 5.01 | 5.89 | 7.04 | 19.81 | 22.36 | 24.74 | 27.69 |

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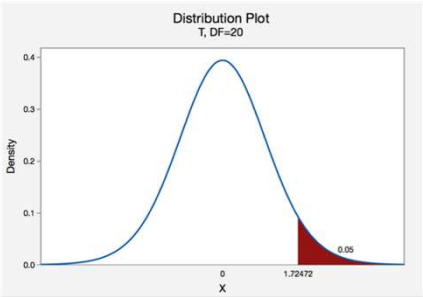
## t-Student Pdf and Probability Table

- For large samples ($n > 30$) the $t$-Student converges to the $N(0,1)$.

| df | 0.40 | 0.25 | 0.10 | 0.05 | 0.025 | 0.01 | 0.005 | 0.001 | .0005 |
|---|---|---|---|---|---|---|---|---|---|
| ... | ... | ... | ... | ... | ... | ... | ... | ... | ... |
| 18 | 0.257 | 0.688 | 1.330 | 1.734 | 2.101 | 2.552 | 2.878 | 3.610 | 3.922 |
| 19 | 0.257 | 0.688 | 1.328 | 1.729 | 2.093 | 2.539 | 2.861 | 3.579 | 3.883 |
| 20 | 0.257 | 0.687 | 1.325 | 1.725 | 2.086 | 2.528 | 2.845 | 3.552 | 3.850 |
| 21 | 0.257 | 0.686 | 1.323 | 1.721 | 2.080 | 2.518 | 2.831 | 3.527 | 3.819 |

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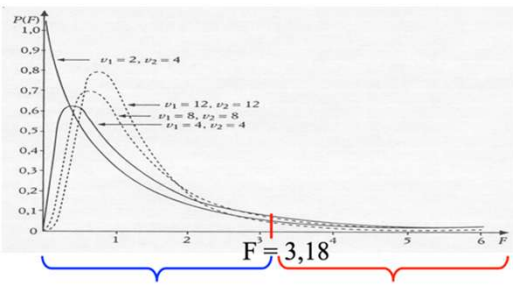
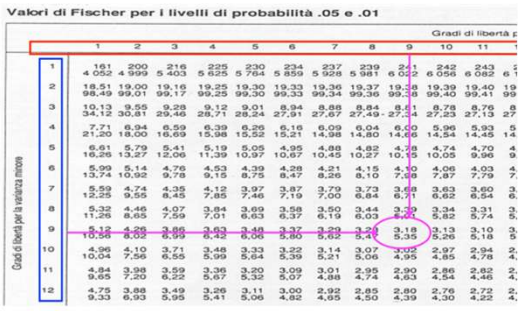
## F-Snedecor-Fisher Pdf and Probability Table

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